The Æther-flow Interpretation of Relativity

First Nonminimal Candidate Branch: Derivative-Mixing Completion of the S/Q Sector

Abstract

The minimal stable-sign candidate action of the derivational bridge has now been shown to possess no nondegenerate flat-background exact-closure branch. The next task is therefore not to restate the benchmark theory, but to define the first nonminimal candidate branch worth testing.

This note performs that branch-selection step in a disciplined way. It first states the admissibility criteria any new candidate branch must satisfy if it is to remain answerable to the exact-closure benchmark of Æther-flow Interpretation of Relativity. It then compares the smallest nonminimal completion families available after the no-go result and selects the derivative-mixing completion of the S/Q sector as the first branch to test. In that branch one adds the operator

$$\Delta\mathcal{L}_\rho = -\rho_I P^{AB}\partial_A S\, D_B Q^I$$

to the minimal auxiliary sector while leaving the bridge metric, the Einsteinian benchmark, and universal matter coupling unchanged.

At quadratic order this completion changes the low-momentum scalar cancellation problem from the minimal positive-sum form to a generalized operator

$$\mathcal{B}_\rho(k^2) = \kappa_S k^2 + \mu_S^2 - \big(\mu_{SI} + \rho_I k^2\big)\big(k^2\delta + \widehat{M}\big)^{-1\,IJ}\big(\mu_{SJ} + \rho_J k^2\big),$$

so that

$$\mathcal{B}_{\rho,2} = \kappa_S + \mu_{SI}\widehat{M}^{-2\,IJ}\mu_{SJ} - 2\mu_{SI}\widehat{M}^{-1\,IJ}\rho_J.$$

The minimal no-go theorem therefore no longer applies in its original form. This note does not prove that the derivative-mixing branch succeeds. It defines the first nonminimal branch cleanly enough that the next quadratic consistency pass can test it directly.

1. Introduction

The active manuscript sequence of Æther-flow Interpretation of Relativity already contains the exact-closure theory statement, its effective dynamics, its consistency analysis, its relativistic recovery statement, its flow-geometry interpretation, the substrate-kinematics bridge manuscript, the coefficient-level linearized consistency pass, and the minimal stable-sign no-go result. The scientific situation is therefore sharper than before. The minimal stable-sign action is no longer an unresolved maybe. It is obstructed as a nondegenerate exact-closure branch.

That shifts the derivational program into a new stage. If the program is to continue beyond the exact-closure benchmark, it must now nominate a concrete nonminimal branch and make clear why that branch is worth testing before any further calculations are undertaken. The present note performs exactly that task.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The scope is deliberately narrow. This note does not claim that the new branch already works. It does not claim a completed Einsteinian derivation. It defines the first nonminimal candidate branch cleanly enough that the next quadratic and curved-background tests can be posed without ambiguity.

2. Admissibility Criteria for a New Candidate Branch

Definition 1 (Admissible post-no-go candidate branch)**.** *A post-no-go candidate branch is admissible for the active derivational program only if it satisfies all of the following.*

1. ***Exact-closure answerability.*** *The branch must retain Æther-flow Interpretation of Relativity as the exact benchmark and must possess a well-defined limit in which the observer-accessible effective theory contracts back to the unique Einsteinian metric sector.*

2. ***Single-metric matter coupling.*** *Matter and light must continue to couple through the bridge metric alone, with no unsuppressed direct low-energy coupling to extra substrate variables.*

3. ***Health discipline.*** *The branch must be testable under sign, ghost, gradient, and mode-control conditions at least as strong as those imposed on the minimal stable-sign branch.*

4. ***Minimal conceptual drift.*** *The new structure should change only the part of the auxiliary sector needed to evade the established obstruction and should not rewrite the public theory claim.*

5. ***Computable next test.*** *The modified branch must admit an immediate flat-background quadratic consistency pass rather than only a speculative verbal motivation.*

Remark 1. *These criteria are methodological, not yet success conditions. They determine whether a proposed branch deserves to enter the manuscript line as a real candidate worth testing.*

3. Smallest Nonminimal Families Worth Considering

After the minimal stable-sign no-go result, the smallest nonminimal completion families worth considering are the following.

1. **Derivative mixing in the S/Q sector.** Add operators that mix the spatial derivatives of S and Q^I while leaving the bridge metric and matter route unchanged.

2. **Constraint-assisted U-sector completion.** Add new auxiliary structure in the longitudinal U-sector so that the scalar cancellation problem is altered through constraint dynamics rather than through direct sign changes.

3. **Higher-derivative completion.** Add operators that postpone the leading scalar obstruction to higher derivative order while keeping the low-energy limit controlled.

The first family is the correct initial target for three reasons. First, it is the smallest local deformation of the auxiliary sector that directly changes the low-momentum scalar coefficient responsible for the no-go result. Second, it does so without modifying the bridge metric, the universal matter-coupling route, or the public exact-closure benchmark. Third, it remains immediately computable at the same quadratic level as the existing linearized-consistency manuscript.

Proposition 1 (First branch-selection result)**.** *The first nonminimal candidate branch worth testing after the minimal stable-sign no-go result is the derivative-mixing completion of the S/Q sector.*

Proof. Among the three families above, only the derivative-mixing family alters the established scalar obstruction at leading low-momentum order while keeping the bridge metric, the matter route, and the benchmark claim unchanged and still allowing an immediate coefficient-level quadratic calculation. It therefore best satisfies the admissibility criteria with the smallest conceptual drift. □

4. Derivative-Mixing Branch Definition

The minimal candidate action is retained, but the auxiliary sector is extended by

$$\Delta\mathcal{L}_\rho = -\rho_I P^{AB}\partial_A S\, D_B Q^I, \tag{1}$$

where ρ_I is a constant coefficient vector on the chosen background branch.

The resulting candidate auxiliary sector is

$$\mathcal{L}_{\text{aux}}^{(\rho)} = \mathcal{L}_{\text{aux}} + \Delta\mathcal{L}_\rho. \tag{2}$$

Definition 2 (Derivative-mixing branch)**.** *The derivative-mixing branch is the branch of the candidate derivational program defined by the action obtained from the minimal bridge action after replacing $\mathcal{L}_{\text{aux}}$ with $\mathcal{L}_{\text{aux}}^{(\rho)}$.*

Remark 2. *The bridge metric $\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B$, the Einsteinian benchmark, and the universal matter-coupling route through $S_{\text{matter}}[\Xi^*\mathfrak{g}, \psi]$ remain unchanged on this branch. Only the auxiliary S/Q sector is modified.*

4.1. Exact-Closure Reversion Limit

The intended reversion limit is

$$\rho_I \to 0. \tag{3}$$

In that limit the derivative-mixing branch contracts exactly back to the minimal candidate action. The exact-closure benchmark of Æther-flow Interpretation of Relativity is therefore preserved as the branch-selection floor rather than rewritten.

5. Quadratic Scalar Sector of the New Branch

On the same admissible homogeneous background used in the minimal quadratic analysis, (1) contributes at quadratic order

$$\Delta\mathcal{L}_{\rho}^{(2)} = -\rho_I\,\partial_i\sigma\,\partial_i q^I. \tag{4}$$

The scalar Q-sector equation is correspondingly modified from the minimal form to

$$\mathcal{A}_{IJ}(-\nabla^2)\,q^J = -(\mu_{SI} + \rho_I(-\nabla^2))\sigma, \tag{5}$$

where

$$\mathcal{A}_{IJ}(-\nabla^2) \equiv (-\nabla^2)\delta_{IJ} + \widehat{M}_{IJ}. \tag{6}$$

Thus

$$q^I = -\mathcal{A}^{-1\,IJ}(-\nabla^2)\,(\mu_{SJ} + \rho_J(-\nabla^2))\sigma. \tag{7}$$

Substituting (7) back into the scalar sector yields

$$\mathcal{L}_{\sigma,\mathrm{eff}}^{(2)} = -\frac{m_S^2}{2}\big(\partial_0\sigma - \sigma_0\phi\big)^2 - \frac{1}{2}\sigma\,\mathcal{B}_\rho(-\nabla^2)\,\sigma, \tag{8}$$

with generalized scalar operator

$$\mathcal{B}_\rho(-\nabla^2) = \kappa_S(-\nabla^2) + \mu_S^2 - \big(\mu_{SI} + \rho_I(-\nabla^2)\big)\mathcal{A}^{-1\,IJ}(-\nabla^2)\big(\mu_{SJ} + \rho_J(-\nabla^2)\big). \tag{9}$$

In Fourier space,

$$\mathcal{B}_\rho(k^2) = \kappa_S k^2 + \mu_S^2 - \big(\mu_{SI} + \rho_I k^2\big)\big(k^2\delta + \widehat{M}\big)^{-1\,IJ}\big(\mu_{SJ} + \rho_J k^2\big). \tag{10}$$

Expanding at small k,

$$\mathcal{B}_\rho(k^2) = \mathcal{B}_{\rho,0} + \mathcal{B}_{\rho,2}k^2 + \mathcal{O}(k^4), \tag{11}$$

with

$$\mathcal{B}_{\rho,0} = \mu_S^2 - \mu_{SI}\widehat{M}^{-1\,IJ}\mu_{SJ}, \tag{12}$$

$$\mathcal{B}_{\rho,2} = \kappa_S + \mu_{SI}\widehat{M}^{-2\,IJ}\mu_{SJ} - 2\mu_{SI}\widehat{M}^{-1\,IJ}\rho_J. \tag{13}$$

Proposition 2 (Why the minimal no-go theorem no longer applies directly)**.** *On the derivative-mixing branch, the low-momentum coefficient $\mathcal{B}_{\rho,2}$ is no longer the sum of manifestly nonnegative terms. The new cross term*

$$-2\mu_{SI}\widehat{M}^{-1\,IJ}\rho_J \tag{14}$$

can compete with the positive minimal contributions without requiring $\kappa_S < 0$ or an indefinite $\widehat{M}_{IJ}$.

Proof. Equation (13) differs from the minimal coefficient exactly by the cross term (14). That term has no fixed positive sign. Therefore the minimal positive-sum argument used in the stable-sign no-go theorem does not transfer unchanged to the derivative-mixing branch. □

Remark 3. *This is not yet a viability proof. It is only the reason the derivative-mixing branch deserves to replace the minimal action as the next branch to test.*

6. Health Conditions That Must Still Be Passed

The derivative-mixing branch is not automatically admissible merely because it escapes the old algebraic obstruction. It must still pass its own health gates.

The first local gate is the spatial-gradient block of the (σ, q^I) sector. At quadratic order that block is

$$\mathbb{K}_{\text{grad}} = \begin{pmatrix} \kappa_S & \rho_J \\ \rho_I & \delta_{IJ} \end{pmatrix}. \tag{15}$$

A basic stable-sign requirement is therefore

$$\mathbb{K}_{\text{grad}} \succeq 0. \tag{16}$$

Equivalently, because the Q-gradient block is the identity in the present branch choice,

$$\kappa_S - \rho_I \delta^{IJ} \rho_J \geq 0. \tag{17}$$

Remark 4. *Condition* (17) *keeps the derivative-mixing branch inside the same sign-discipline philosophy as the minimal stable-sign branch while still permitting nonzero* ρ_I.

Further tests remain mandatory:

1. ghost and kinetic-sign control in the full quadratic sector
2. absence of unsuppressed extra observer-accessible poles
3. preservation of single-metric matter coupling
4. compatibility with local relativistic recovery and the exact-closure benchmark
5. curved-background continuation beyond the Minkowski patch

7. Immediate Research Tasks for This Branch

Now that the branch has been chosen and written explicitly, the next calculations are fixed.

1. derive the full quadratic auxiliary Lagrangian of the derivative-mixing branch, including any corrections to the longitudinal U-sector elimination
2. determine the exact cancellation conditions replacing the minimal $\mathcal{B}_0 = \mathcal{B}_2 = 0$ test
3. check whether a nondegenerate flat-background exact-closure branch exists subject to (16)
4. test whether the branch retains a healthy spectrum and controlled matter coupling
5. only after that, extend the analysis to slowly varying and then global admissible backgrounds

8. Conclusion

The minimal stable-sign action no longer serves as a serious open candidate branch. After the no-go result, the derivational program needed a new branch worth testing, not a repetition of the benchmark theory. This note provides that next step.

The first branch worth testing is the derivative-mixing completion of the S/Q sector. It is the smallest nonminimal deformation that changes the leading scalar cancellation problem while preserving the bridge metric, the matter route, and the exact-closure benchmark of Æther-flow Interpretation of Relativity. Its generalized coefficient $\mathcal{B}_{\rho,2}$ contains a new cross term that removes the old minimal positive-sum obstruction.

That is enough to promote the branch into the next calculation stage, but not enough to declare success. The derivative-mixing branch now has to survive the same discipline demanded elsewhere in the active sequence: explicit quadratic coefficient control, health of the observer-accessible spectrum, and eventual curved-background continuation. Until it does, exact closure remains the scientific benchmark and reversion theory of the project.